

Chapter 2

The Determinants of Exchange Rates

Chapter Outline

- Exchange rate basics
- Calculating exchange rates
- Main factors affecting exchange rates
- The role of central banks in currency values
- Fundamentals of central bank intervention
- The Equilibrium approach to exchange rates

Exchange Rate Basics

- What are exchange rates?
- Exchange rate notation
- The base currency and the reference currency
- Exchange rate systems
- Computing prices of goods in foreign countries

What are Exchange Rates?

- Foreign exchange is often denoted as “FX.”
- What is an exchange rate?
 - The price of one nation’s currency in terms of another nation’s currency.
 - Represents the cost of exchanging one currency for the other (or purchasing one currency using the other).
- Determined by supply and demand
- Exchange rates can be traded on foreign exchange (“forex”) markets for spot (immediate) or forward (future) delivery.

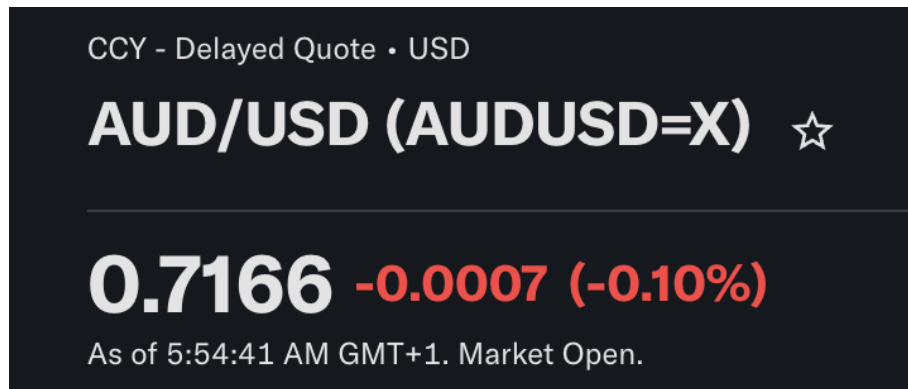
Examples of Prices

1 loaf of bread = \$3.95
1 gallon of milk = \$3.79
1 pound of beef = \$14.99
1 British pound = \$1.65
1 Swiss franc = \$1.17

Exchange rates
are simply prices
of foreign
currencies

Exchange rate notation: market convention

- Exchange rates can be quoted using different conventions.
- In **market convention**: **A/B** means the number of units of currency **B** per 1 unit of currency **A**.
 - Under this convention: A = base currency and B = reference currency.
 - Example: AUD/USD = 0.7166. This means: 1 AUD = USD 0.7166.
 - AUD is the *base currency* and USD is the *reference currency*.



Source: Yahoo Finance. <https://finance.yahoo.com/quote/AUDUSD=X/>

Exchange rate notation: Textbook convention

- However, a number of publications including textbooks such as the main **textbook** in this course (Shapiro) use a different notation.
- In the **textbook**: **A/B** means the number of units of currency A per 1 unit of currency B.
 - Under this convention: B = base currency and A = reference currency.
 - Example: AUD/USD = 1.3955. This means: 1 USD = AUD 1.3955
 - This can also be written as AUD 1.3955/USD.
 - USD is the *base currency*, and AUD is the *reference currency*.
- **IMPORTANT:** to avoid confusion, this course follows the **textbook notation**. That is: **A/B** means the number of units of A per 1 unit of B.

Exchange rate notation: Textbook convention (Cont)

- The textbook also often uses the term *home (domestic) currency* and *foreign currency*.
- Because the textbook is written from a **US** perspective, it usually treats:
USD = domestic (home) currency
Non-USD currencies = foreign currencies
- The textbook often writes: **Home currency/Foreign currency**. This means:
number of home currency units per 1 unit of foreign currency.
 - In other words, foreign currency = base currency and home currency = reference currency.
 - Example: USD/EUR=1.20, or USD 1.20/EUR. This means: 1 EUR = USD 1.20.
 - EUR = base currency & USD = reference currency

American vs European Quotations of currencies

1. PLEASE NOTE THAT STUDENTS ARE NOT REQUIRED TO KNOW THE FOLLOWING DEFINITIONS FOR THEIR EXAMS. HOWEVER, WE ARE COVERING THEM FOR INFORMATIONAL PURPOSES ONLY, AND IT IS BEST NOT TO FOCUS ON THEM, AS WE DO NOT REFER TO THEM IN THIS COURSE
2. The American Currency Quotation (defined as direct quote) shows how many USD it takes to purchase one unit of foreign currency.
3. On the other hand, the European Currency Quotation (defined as indirect quote) shows how much foreign currency is needed to purchase one unit of USD.

Why are many publications, including textbooks, exchange rate quotes against the USD?

Many publications and textbooks examples quote currencies against the US dollar because USD is widely used in international trade and financial markets. Reasons for them include:

Global Dominance: Established by the Bretton Woods Agreement in 1944, the USD remains the preferred **reserve currency**. Central banks and financial institutions hold vast amounts of their reserves in U.S. dollars.

International Trade: The vast majority of global cross-border trade (such as oil and commodities) is priced and settled in USD.

Liquidity and Stability: Because the U.S. has the world's largest economy, the USD is highly liquid, allowing large transactions to take place smoothly and with lower conversion costs.

\$1 = Mex\$5.33 Mexico Peso

\$1 = Z\$38 Zimbabwe

\$1= ¥ 100 (Japanese Yen)

Cross Rates

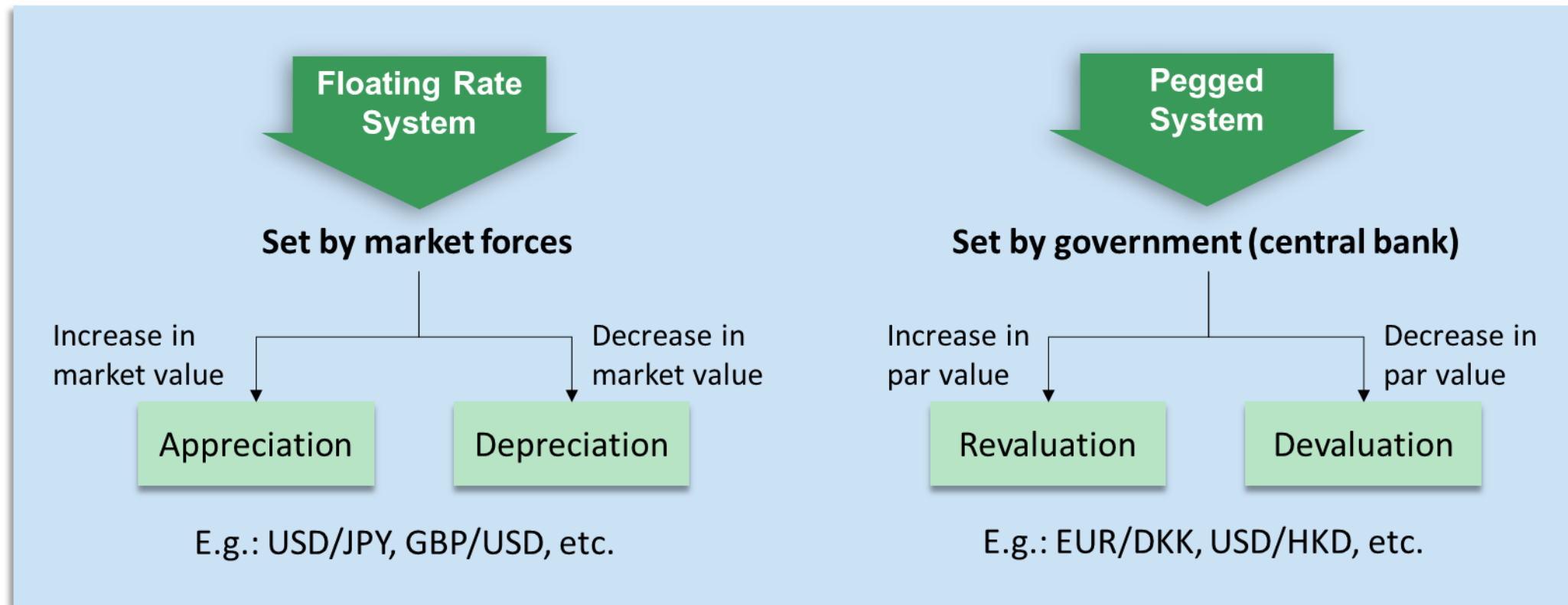
1. A cross rate is the exchange rate between two currencies calculated using a third currency. It is used when the two currencies aren't commonly traded directly against each other.
2. How are cross rates calculated?
3. Compare the exchange rate of each currency in the pair against a third currency, often the US dollar
2. Use the exchange rates against the third currency to calculate the exchange rate between the two currencies

Examples of Cross Rates

Examples of cross rates:

- EUR/AUD (the exchange rate between the euro and the Australian dollar)
- EUR/GBP (the exchange rate between the euro and the British pound)
- CHF/JPY (the exchange rate between the Swiss franc and the Japanese yen)
- AUD/NZD (the exchange rate between the Australian dollar and the New Zealand dollar)

Exchange Rate Systems within the International Monetary System



Equilibrium Exchange Rates

2.1 SETTING THE EQUILIBRIUM

A. **The exchange rate**

is the price of one unit of foreign currency expressed as a certain price in the US dollar (home currency).

For example, \$1.20/€ means 1 Euro is worth \$1.20.

Computing Prices of Goods in Foreign Countries

- A BMW costs €62,500 in Germany.
- The current exchange rate is \$1.20/€
- How much does the BMW cost in US dollars?
- Answer: **$62,500 \times 1.20 = \$75,000$.**



Equilibrium Exchange Rates

1. When Americans Purchase German Goods:

Foreign Currency Demanded derived from the demand for foreign country's goods, services, and financial assets.

e.g. The demand for German cars by Americans

Equilibrium Exchange Rates

2. Foreign Currency Supply:

- a. derived from the foreign country's demand for local (i.e. US) goods.

- b. Foreigners must convert their currency to purchase US goods.

e.g. German demand for US goods means Germans convert € to US \$ in order to buy the US goods.

Supply and Demand (1 of 3)

- **Demand for a currency**

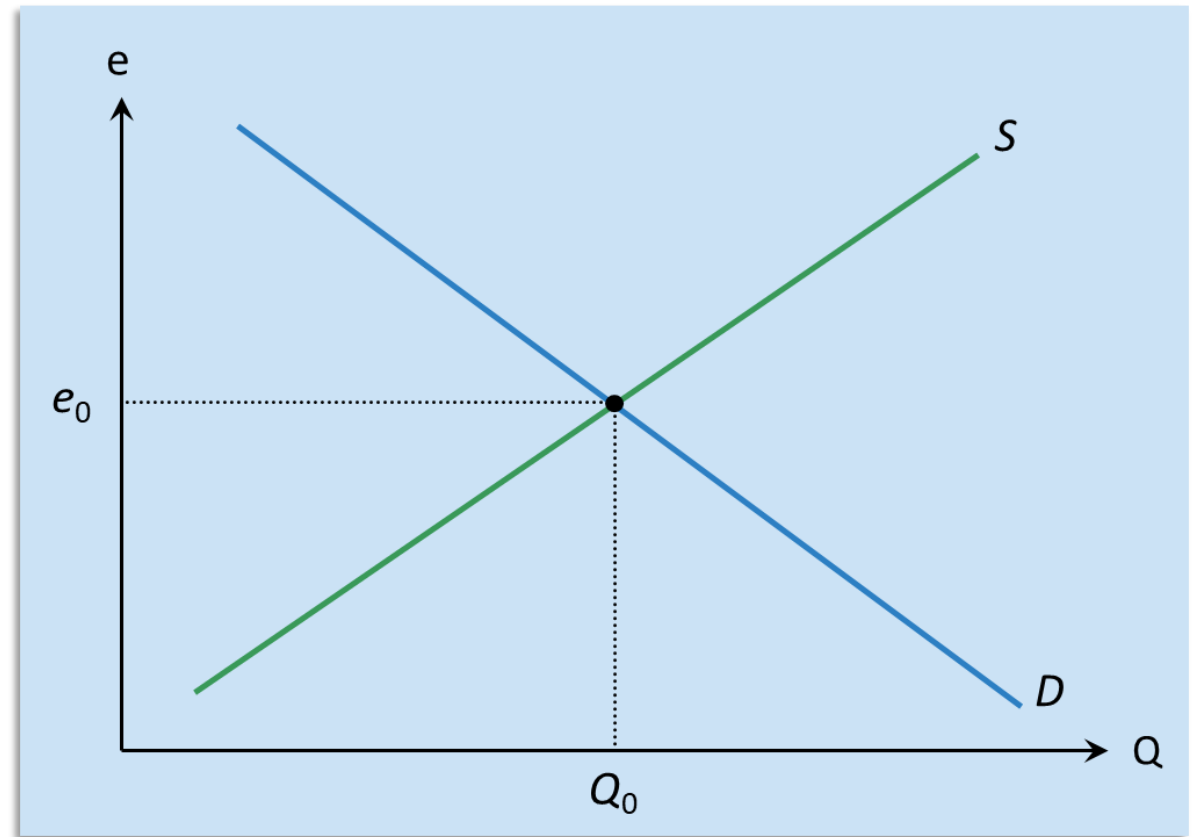
- Demand for a currency is a function of the demand for foreign goods denominated in that currency. E.g., US demand for Eurozone goods increases demand for euros to pay for those goods.
- As demand for euros increases, the value of the dollar falls against the euro.
- As the value of the dollar falls against the euro, Americans demand fewer Eurozone goods, services, and assets.

- **Supply of a currency**

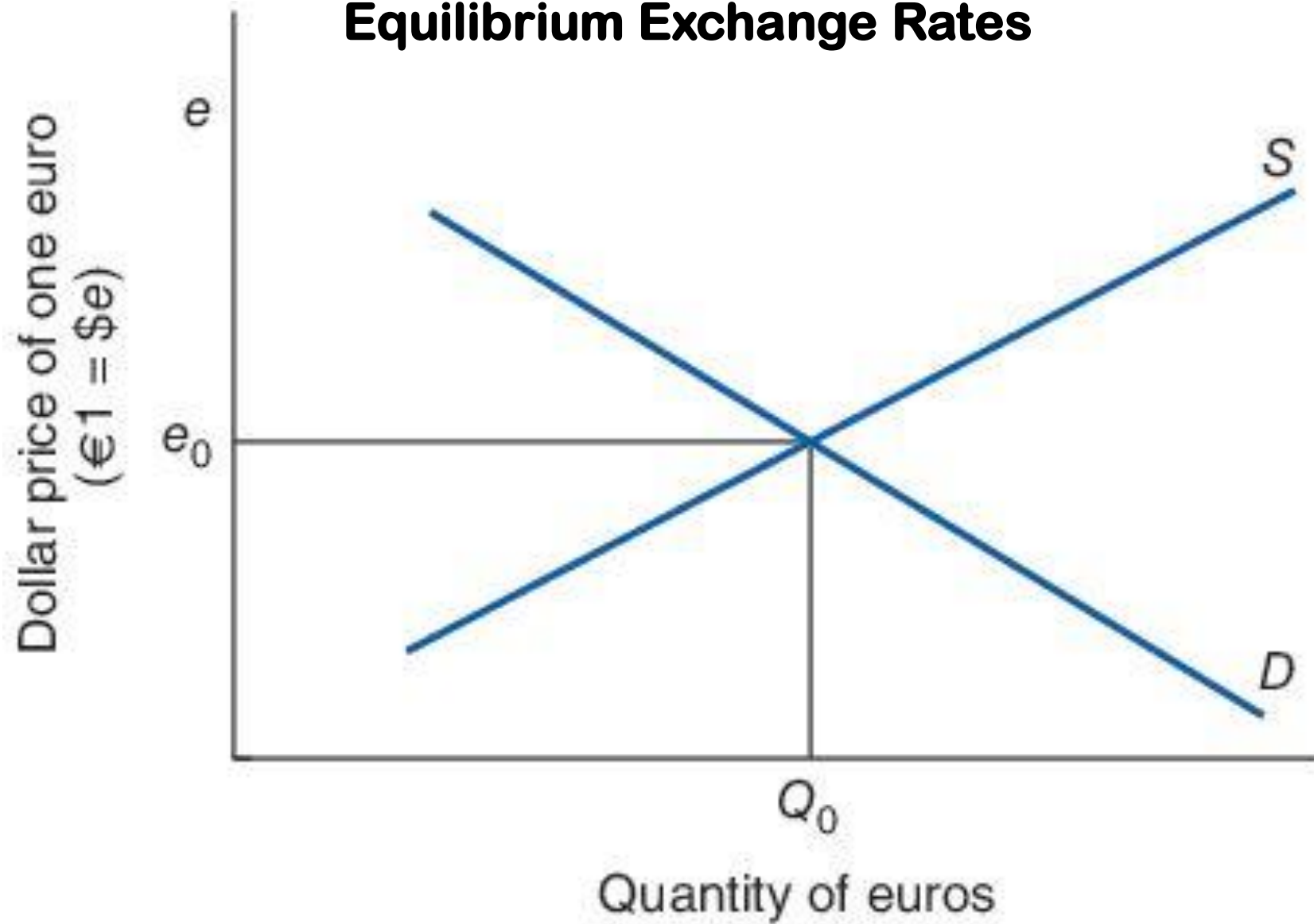
- Supply of euros is a function of Eurozone demand for U.S. goods.
- Eurozone consumers must buy dollars to buy U.S. goods.
- As the value of the euro increases against the dollar, increased Eurozone demand for U.S. goods increases demand for dollars, which increases the amount of euros supplied.

Supply and Demand (2 of 3)

- The exchange rate is the price, e_0 , that equates supply with demand.



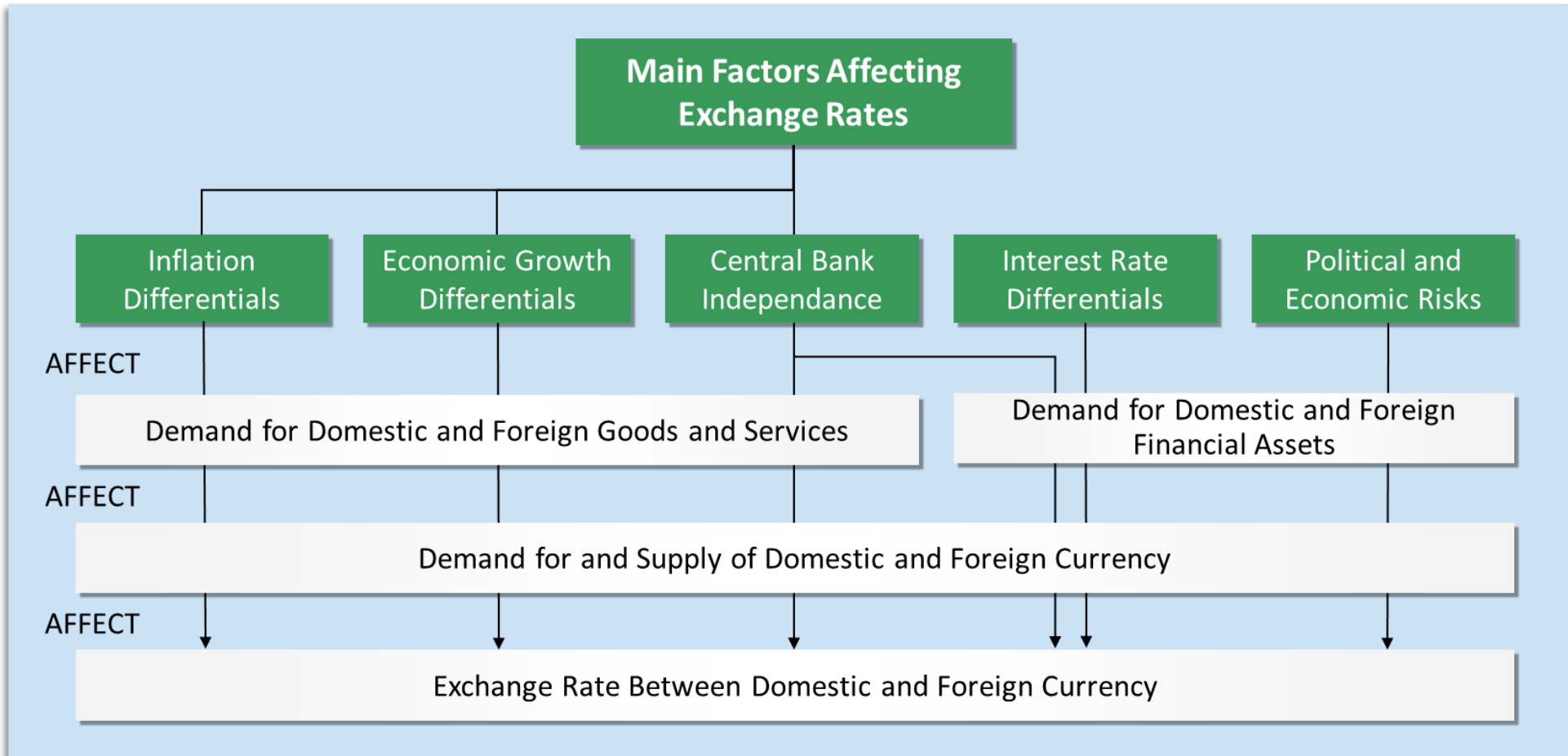
Equilibrium Exchange Rates



Main Factors Affecting Exchange Rates

- How certain factors affect exchange rates
- Supply and demand
- Inflation factor
- Interest rate factor
- Economic growth factor
- Political and economic risk factor
- Summary

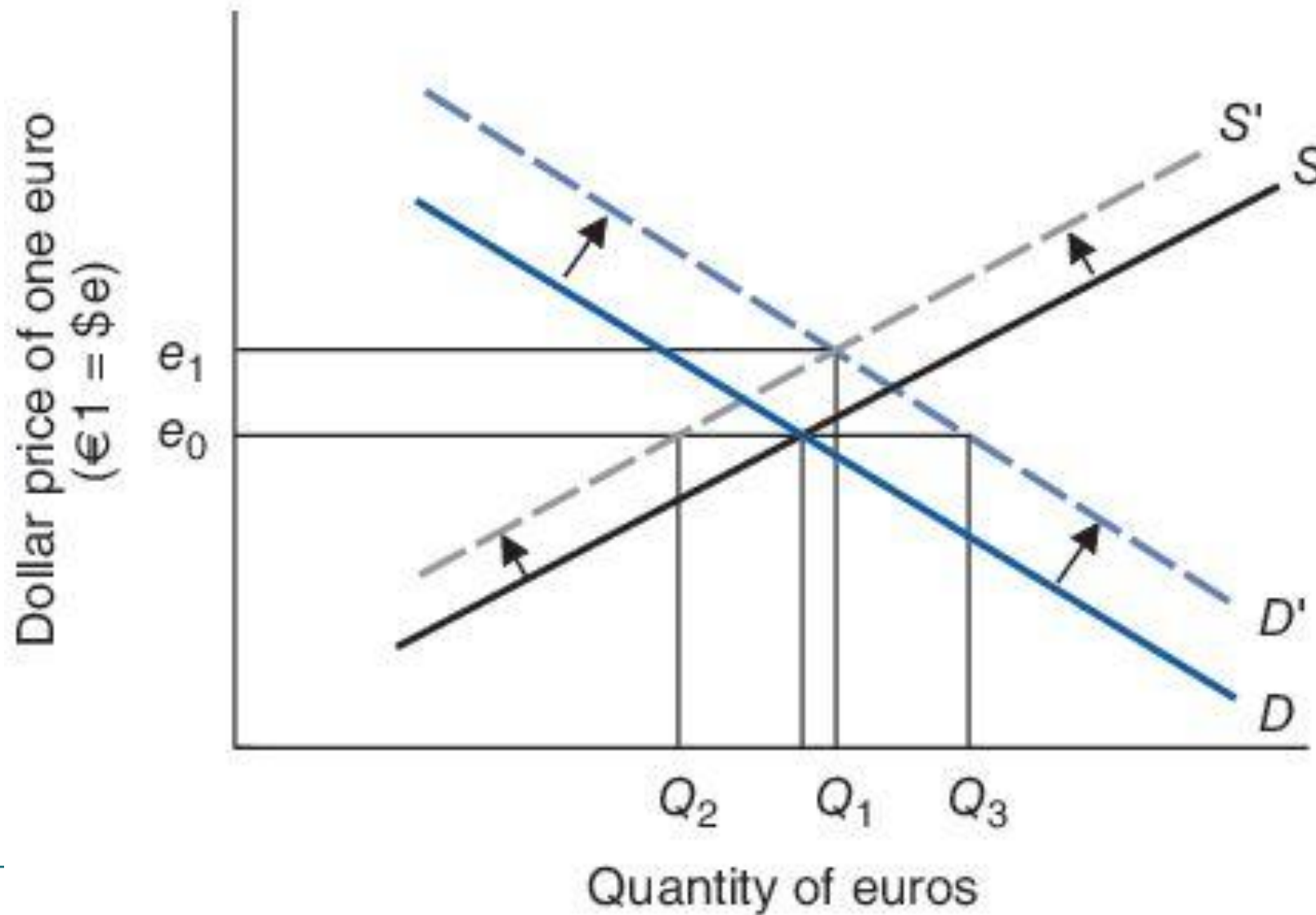
How the Main Factors Affect Exchange Rates



Consequence of higher inflation in the US compared to Eurozone

- European buyers purchase fewer US products due to high US inflation. This means that high US inflation leads to a decrease in the supply of Euros, as fewer goods are purchased from the US. The supply curve for Euros shifts to the left.
- At the same time, higher prices in the US will lead American consumers to substitute Eurozone imports for US products, resulting in a shift of the Euro demand curve to the right. This will lead to a new exchange rate where the US dollar depreciates against the Euro, or the Euro appreciates against the US dollar. (When prices are high in America, consumers face a higher cost of living. To stretch their budgets, they naturally trade down from premium, imported Eurozone goods (which often include high shipping, tariff, and currency conversion markups) to more affordable domestic alternatives).

Impact of U.S. Inflation on the Equilibrium Exchange Rate



The consequences of higher inflation in the US vs the Euro Zone on the exchange rate between the US dollar and the Euro

When US inflation rises faster than Eurozone inflation, the **US Dollar (USD)** **generally weakens** against the **Euro (EUR)**. Consequently, Eurozone **exports to the US become more expensive** and less competitive, while **imports from the US become cheaper** and more attractive to Eurozone buyers.

1. Eurozone Exports and Imports

Exports to the US: Because the Euro strengthens against the dollar, Euro-priced goods cost more when converted to US dollars. This tends to reduce the volume of Eurozone exports to the US as they become more expensive for US consumers.

The consequences of higher inflation in the US vs the Euro Zone on the exchange rate between the US dollar and the Euro

Imports from the US: A stronger Euro increases the purchasing power of Eurozone buyers. US products become comparatively cheaper to buy in Europe, which generally drives an increase in Eurozone imports from the United States.

Definition of a real interest rate

A real interest rate is a stated interest rate with the rate of inflation removed from it. It reflects the real cost of funds to a borrower and the real yield to a lender or to an investor.

A real interest rate relates to the rate of time preference for current goods over future goods. It is calculated as the difference between the nominal interest rate and inflation rate.

Real interest rate = nominal interest rate - rate of inflation (expected or actual).

The impact of an increase in US interest rates on the US dollar vs Euro currency

An increase in US interest rates attracts global capital by offering higher yields, causing the US dollar to appreciate. For euro zone investors, this shift generally strengthens the **USD** against the **Euro**, creating the potential for higher returns on US securities but also introducing a significant **currency risk**.

1. Consequences for Euro Zone Investors

Attractive Yields: Higher US interest rates directly translate to better yields on fixed-income US securities (like Treasury bonds) compared to European equivalents.

The impact of an increase in US interest rates on the US dollar vs Euro currency

Currency Gains (or Losses): When Euro investors buy US assets, they must first exchange their Euros for US dollars. If the USD strengthens against the Euro, these investors benefit from a double gain: the initial yield from the security plus a conversion gain when profits are translated back to Euros.

2. Impact on Currency Values Appreciation of the US Dollar: As European (and global) investors sell Euros to purchase higher-yielding US securities, the demand for the US dollar spikes. This increased demand naturally pushes up the value of the US dollar in the foreign exchange market.

Depreciation of the Euro: The reverse is true for the Euro. The mass selling of Euros to fund these US investments creates an oversupply of the currency, causing the Euro to depreciate or weaken against the dollar

Economic Growth Factor

- Investors prefer to invest in high growth markets.
 - Countries with fast-growing GDP have more investment opportunities, attract more capital, and have higher currency values.
 - Economic growth is good for nation's currency.
- High currency values may have a feedback effect on economic growth.
 - Higher currency values make imports cheaper and hurt exports.
 - Since the trade balance is a component of GDP, high currency values can negatively affect economic growth.

Political and Economic Risk Factor

- Investors are risk-averse (like returns, dislike risk that may affect returns).



- A currency's riskiness depends on a nation's economic and political policies.
 - The more economically and politically stable a nation is, the higher the demand for its currency will be.
- In times of turmoil, markets flock to stable currencies (from economically and politically stable countries) such as the US dollar or the Swiss franc.

Equilibrium Exchange Rates

To ensure students become familiar with the use of exchange rate notation in this course, including the PPT slides for Chapter 2, they are encouraged to use only the following PPT slides for exchange rate notation rather than reading other sources, including the actual textbook of this course, for PPT slides 32 to 43, when we present numerical examples.

Home Currency Depreciation

- a. Foreign currency more valuable than the home currency
- b. Conversely, the foreign currency's value has appreciated against the home currency

Equilibrium Exchange Rates

- Using the **Textbook convention (see Slide 6)**, for an FX quote **A/B**:
 - **A** = reference currency & **B** = base currency
 - So: **A/B = e** means 1 unit of **B** = **x** units of **A**
- Also, recall that the textbook often quotes: Home currency/Foreign currency
 - In this case: Home (domestic) currency = reference currency & foreign currency = base currency.

% change in the value of the base currency **B:**

$$= (e_1 - e_0) / e_0$$

where e_0 = old exchange rate

e_1 = new exchange rate

If $e_1 > e_0$, the base currency appreciates;

if $e_1 < e_0$, the base currency depreciates.

Equilibrium Exchange Rates

% change in the value of the reference currency [A](#):

$$= (e_0 - e_1) / e_1$$

where e_0 = old exchange rate
 e_1 = new exchange rate

If $e_1 > e_0$, the reference currency depreciates

If $e_1 < e_0$, the reference currency appreciates

Calculating Exchange Rates (1 of 4)

- Depending on the current value of the euro relative to the dollar, the amount of euro appreciation or depreciation (that could be, for instance, due to higher inflation in the US vs Eurozone) is computed as the fractional increase or decrease in the dollar value of the euro.
- For example, if the **\$/€** exchange rate goes from €1 = \$0.93 to €1 = \$1.09, the euro is said to have appreciated by the change in its dollar value, which is: $(1.09 - 0.93)/0.93 = 17.20\%$. We obtain a general formula:

$$\text{Amount of euro appreciation (depreciation)} = \frac{\text{New dollar value of euro} - \text{Old dollar value of euro}}{\text{Old dollar value of euro}}$$

Calculating Exchange Rates (2 of 4)

$$\% \Delta \textit{Base currency (EUR)} = \frac{e_1 - e_0}{e_0} \quad \textbf{(Equation 1)}$$

- Substituting in the numbers from the previous example (with $e_0 = \$0.93$ and $e_1 = \$1.09$) yields **the 17.20% euro** appreciation.
- Alternatively, we can calculate the change in the euro value of the dollar. We can do this by recognizing that if e equals the dollar value of a euro (dollars per euro), then the euro value of a dollar (euros per dollar) must be the reciprocal, or $1/e$.

Calculating Exchange Rates (3 of 4)

- For example, if the euro is worth \$0.93, then the dollar must be worth €1.075 (or $1/0.93$) .
- The change in the euro value of the dollar between time 0 and time 1 equals $1/e_1 - 1/e_0$. In percentage terms, the dollar is said to have depreciated (appreciated) against the euro by the fractional decrease (increase) in the euro value of the dollar:

$$\% \Delta \text{Reference Currency (USD)} = \frac{\frac{1}{e_1} - \frac{1}{e_0}}{\frac{1}{e_0}} = \frac{e_0 - e_1}{e_1} \quad \text{(Equation 2)}$$

Calculating Exchange Rates (4 of 4)

- Employing Equation 2, we can find the increase in the euro exchange rate from \$0.93 to \$1.09 to be equivalent to a dollar depreciation of

$$\frac{0.93 - 1.09}{1.09} = -14.68\%$$

- Why is it that if we look at changes in the domestic currency relative to the foreign one, we get a different answer than if we look at the change in the opposite direction?
- **Answer (This issue is not examinable and students should only note it:** The reason the euro appreciation is unequal to the amount of dollar depreciation depends on the fact that the value of one currency is the inverse of the value of the other one. Hence, the percentage change in currency value differs because the base from which the change is measured differs. To dive deeper into how central banks and economists model and measure these exchange rates, refer to resources provided by the [Reserve Bank of Australia](#) or review working papers published by the [National Bureau of Economic Research](#).

Calculating Currency Appreciation and Depreciation (1 of 7)

$$\% \text{ Change in the value of the base currency} = \frac{e_1 - e_0}{e_0}$$

Where e_0 = exchange rate at time 0

e_1 = exchange rate at time 1

Example 1a:

- In 2022, the yen dropped against the dollar (i.e., \$/¥ from \$0.0086878/¥ to \$0.0076482/¥)
- How much did the ¥ depreciate against the \$? $\Rightarrow \frac{0.0076482 - 0.0086878}{0.0086878} = -11.97\%$
- **Explanation:** Using above Equation 1, we calculate the percentage change in the dollar price of the yen by subtracting the exchange rate at time 0 from the exchange rate at time 1, then dividing by the exchange rate at time 0. Since the dollar price of the yen fell from 0.0086878 to 0.0076482, the yen depreciated against the dollar.

Calculating Currency Appreciation and Depreciation (2 of 7)

$$\% \Delta_{\text{Reference currency}} = \frac{\frac{1}{e_1} - \frac{1}{e_0}}{\frac{1}{e_0}} = \frac{e_0 - e_1}{e_1} \quad (\text{Equation 2})$$

Example 1b:

- In 2022, the yen dropped against the dollar (i.e., \$/¥) from \$0.0086878/¥ to \$0.0076482/¥.
- How much did the \$ appreciate against the ¥? $\Rightarrow \frac{1/0.0076482 - 1/0.0086878}{1/0.0086878} = +13.59\%$
- **Explanation:** Using above Equation 2, we calculate the percentage change in the yen price of the dollar by first converting the exchange rates into their reciprocals. We then subtract the exchange rate at time 0 from the exchange rate at time 1 and divide by the exchange rate at time 0. Since the yen price of the dollar increased, the dollar appreciated against the yen.

Calculating Currency Appreciation and Depreciation (3 of 7)

$$\% \text{ Change in the value of the base currency} = \frac{e_1 - e_0}{e_0}$$

Where

e_0 = exchange rate at time 0

e_1 = exchange rate at time 1

Example 2a:

- On June 24, 2016, the British pound dropped against the dollar (i.e., $\$/\pounds$) from \$1.4883/£ to \$1.3694/£.
- How much did the £ depreciate against the \$? $\rightarrow \frac{1.3694 - 1.4883}{1.4883} = -8.00\%$
- **Explanation:** Using above Equation 1, we calculate the percentage change in the dollar price of the pound by subtracting the exchange rate at time 0 from the exchange rate at time 1, then dividing by the exchange rate at time 0. Since the dollar price of the pound fell from 1.4883 to 1.3694, the pound depreciated against the dollar.

Calculating Currency Appreciation and Depreciation (4 of 7)

$$\% \Delta_{\text{reference currency}} = \frac{\frac{1}{e_1} - \frac{1}{e_0}}{\frac{1}{e_0}} = \frac{e_0 - e_1}{e_1} \quad (2)$$

Example 2b:

- On June 24, 2016, the British pound dropped against the dollar (i.e., \$/£) from \$1.4883/£ to \$1.3694/£.
- How much did the dollar appreciate against the £? $\Rightarrow \frac{1/1.3694 - 1/1.4883}{1/1.4883} = +8.68\%$
- Explanation:** Using above Equation 2, we calculate the percentage change in the pound price of the dollar by first converting the exchange rates into their reciprocals. We then subtract the exchange rate at time 0 from the exchange rate at time 1 and divide by the exchange rate at time 0. Since the pound price of the dollar increased, the dollar appreciated against the pound.

Calculating Currency Appreciation and Depreciation (5 of 7)

$$\% \text{ Change in the value of the base currency} = \frac{e_1 - e_0}{e_0}$$

Where

e_0 = exchange rate at time 0

e_1 = exchange rate at time 1

Example 3a:

- From Jan. 2021, to Sept. 2022, the dollar rose against the euro (i.e. \$/€) from \$1.2011/€ to \$0.9802/€.
- How much did the € depreciate against the \$? $\rightarrow \frac{0.9802 - 1.2011}{1.2011} = -18.39\%$
- **Explanation:** Using Equation 1, we calculate the percentage change in the dollar price of the euro by subtracting the exchange rate at time 0 from the exchange rate at time 1, then dividing by the exchange rate at time 0. Since the dollar price of the euro fell from 1.2011 to 0.9802, the euro depreciated against the dollar.

Calculating Currency Appreciation and Depreciation (6 of 7)

$$\% \Delta_{\text{Reference Currency}} = \frac{\frac{1}{e_1} - \frac{1}{e_0}}{\frac{1}{e_0}} = \frac{e_0 - e_1}{e_1} \quad (2)$$

Example 3b:

- From Jan. 2021, to Sept. 2022, the dollar rose against the euro (i.e. \$/€) from \$1.2011/€ to \$0.9802/€.
- How much did the \$ appreciate against the €? $\rightarrow \frac{1/0.9802 - 1/1.2011}{1/1.2011} = +22.54\%$
- **Explanation:** Using above Equation 2, we calculate the percentage change in the euro price of the dollar by first converting the exchange rates into their reciprocals. We then subtract the exchange rate at time 0 from the exchange rate at time 1 and divide by the exchange rate at time 0. Since the euro price of the dollar increased, the dollar appreciated against the euro.

Calculating Currency Appreciation and Depreciation (7 of 7)

$$\% \text{ Change in the value of the base currency} = \frac{e_1 - e_0}{e_0}$$

Where e_0 = exchange rate at time 0

e_1 = exchange rate at time 1

Example 4:

- July 19th, 1985, the Italian lira depreciated by 17% against the dollar. What was the appreciation of the dollar against the lira?

- We are given: $\frac{e_1 - e_0}{e_0} = -17\%$

$$e_1 = -0.17e_0 + e_0 = 0.83e_0$$



- We want: $\frac{e_0 - e_1}{e_1} = ?$

$$\frac{e_0 - 0.83e_0}{0.83e_0} = \frac{0.17}{0.83} = 20.48\%$$

Expectations and the Asset Model of Market Model of Exchange Rate

The Role of Central Banks in Currency Values

- Central bank reputation and currency value
- Central bank independence
- Inflation creates an economic “hangover”
- Currency substitution or dollarization

Central Bank Reputation and Currency Value

- A central bank uses instruments of monetary policy to achieve three primary outcomes:

Price
Stability

Low Interest
Rates

Target
Currency

- Thus, a central bank's actions influence all of the factors that impact currency value.
- Because expectations of central bank behavior are important, reputation is important.
 - If a central bank creates too much money, inflation will occur and currency value will fall. Thus:
 - Currencies of central banks with a favorable record of protecting a currency's purchasing power sell at a premium.
 - Conversely, currencies of central banks with a poor record trade at a discount.

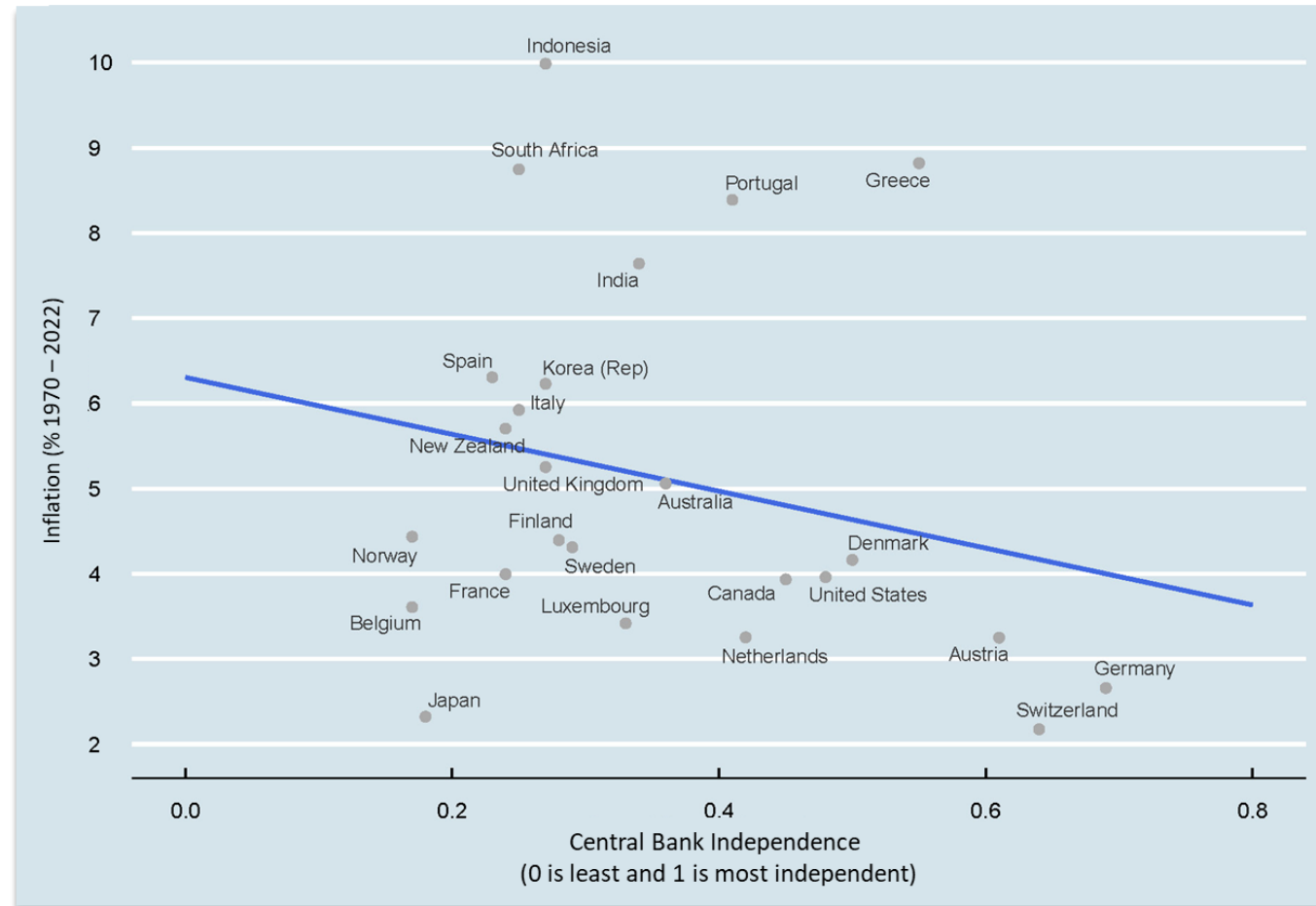
Central Bank Independence (1 of 2)

- Central bank independence is a critical element in determining soundness of monetary policy.
 - The degree to which the bank is operationally and managerially independent from government.
- Central banks adopt mandates for price stability that are verifiable, unambiguous, and enforceability
- The absent of mandates creates the risk of political influence in central bank activities.
- Non-independent central banks may be forced to monetize the deficit by buying government debt with newly created money
 - E.g., In 1920, Germany resorted to inflation to fund WW1 reparations to the Allies.



In 1923, it took 4 banknotes to buy a loaf of bread.

Central Bank Independence (2 of 2)



Fundamentals of Central Bank Intervention

- Real exchange rates and relative competitiveness
- Foreign exchange market intervention

Real Exchange Rates and Relative Competitiveness (1 of 3)

- Home currency appreciation or depreciation a “mixed blessing.”

Home Currency Appreciation (strong US\$)

Advantages	Disadvantages
• Dollar prices of imports fall, benefiting US consumers. • Dollar prices of exports are higher, benefiting US firms with products with low substitution. • Lower import prices put downward pressure on prices of competing US products, reducing domestic inflation. • Cost of foreign investment is lower for US firms and individuals.	• US exports become less competitive in foreign markets. • Domestic competition from lower-priced foreign imports increases. • US loses jobs in the US traded-goods sector. • Higher cost of operating in the US reduces foreign direct investment, slowing job creation by foreign firms.

Real Exchange Rates and Relative Competitiveness (2 of 3)

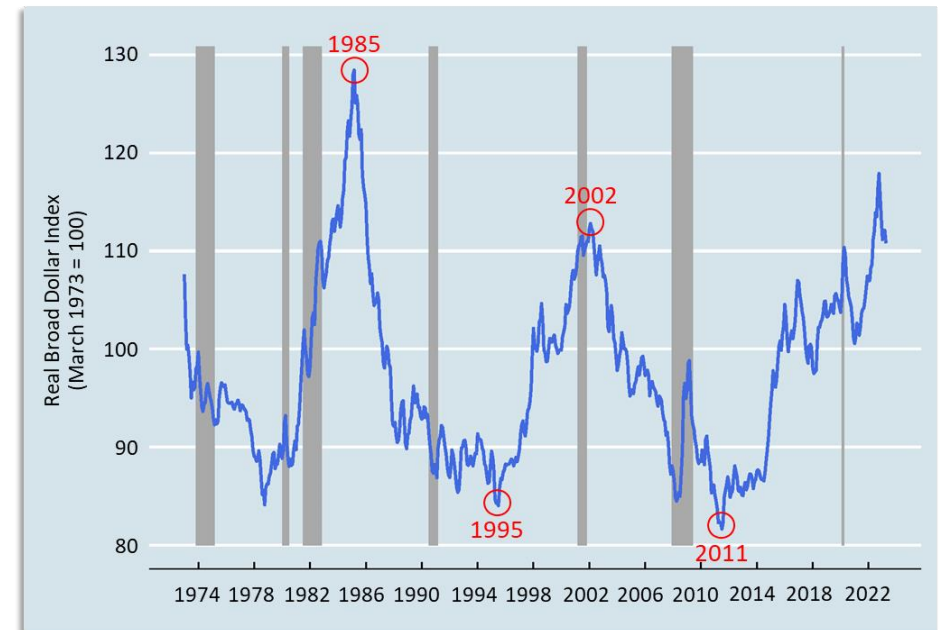
Home Currency Depreciation (weak US\$)

Advantages	Disadvantages
• US exports become more competitive in foreign markets. • US firms are more competitive domestically against higher-priced foreign imports. • US gains jobs in the traded-goods sector. • Lower cost of operating in the US increases foreign direct investment in the US, boosting job creation by foreign firms	• Dollar prices of imports are higher, hurting US consumers. • Dollar prices of exports are lower, hurting firms with products with low substitution. • Reduced price competition from imports leads to price increases by competing US products, increasing US inflation. • Cost of foreign investment is higher for US firms and individuals. • Foreign capital fleeing a weak currency leads to higher US interest rates.

Real Exchange Rates and Relative Competitiveness (3 of 3)

Example: Fluctuating dollar over time

- From its peak in mid-1985, fell by more than 50% during the next few years
 - US companies were more competitive worldwide
 - Concurrently, foreign goods and services were more expensive for Americans.
- Hit a low point in 1995 and then began to strengthen, due largely to reductions in both inflation and the budget deficit, and to strong economic growth.
- After reaching a high in 2002, started falling, due partly to higher budget and trade deficits, eventually reaching its lowest point in 2011 during the Great Recession.



Foreign Exchange Market Intervention (1 of 6)

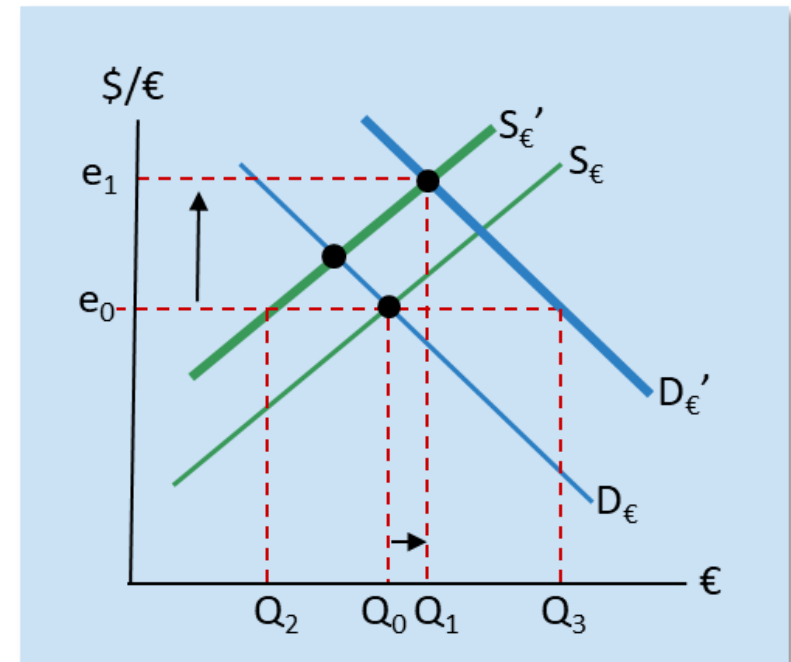
- In this course, we do not cover the following subject matter. However, we need to have some basic understanding of the quantitative Easing policies by the central banks. Foreign exchange market intervention refers to official purchases and sales of foreign exchange made by central banks to influence their currencies.

- Example: Suppose the US and Eurozone decide to maintain the old exchange rate e_0 in the face of the new equilibrium rate e_1 .

Excess demand for euros	Excess supply of dollars (= euro shortage)
$Q_3 - Q_2$ euros	$(Q_3 - Q_2) \times e_0$ dollars

- The Federal Reserve or European Central Bank, or both, would have to intervene in the market to supply this additional quantity of euros (i.e., to buy up the excess supply of dollars).
- Absent intervention:

US balance-of-payments deficit	Eurozone balance-of-payments surplus
$(Q_3 - Q_2) \times e_0$ dollars	$Q_3 - Q_2$ euros



Foreign Exchange Market Intervention (2 of 6)

Two types of market intervention

Unsterilized

Market intervention

- From previous example, \$ supply falls and € supply rises.
- All things equal, an increase/decrease in the money supply will result in more/less inflation.
- Thus, market intervention not only changes the exchange rate, it increases Eurozone inflation while reducing US inflation.
- These changes will also affect interest rates in both countries.

Sterilized

Central bank intervention

(through open-market operations)

- By purchasing treasury securities, the money supply increases.
- By selling treasury securities, the money supply decreases.
- Example: The Federal Reserve purchases enough T-bills to restore the money supply to pre-intervention levels.

Foreign Exchange Market Intervention (3 of 6)

- Central bank (sterilized) intervention is often ineffective.
 - To permanently affect the exchange rate, investors must view domestic and foreign securities as imperfect substitutes.
 - **If they do**, the exchange rate and relative interest rates must change to induce investors to hold the new securities.

E.g.: Bank of Japan sells yen to decrease its value



1. ¥ value falls below what it would have without intervention;
2. Interest rate on ¥ bonds increases above what it would have without intervention;
3. Interest rate on \$ assets falls below what it would have without intervention.

- If investors view the ¥ and \$ as imperfect substitutes, no exchange rate or interest rate changes would be necessary to convince them to hold them.
- Thus, sterilized intervention would be ineffective.

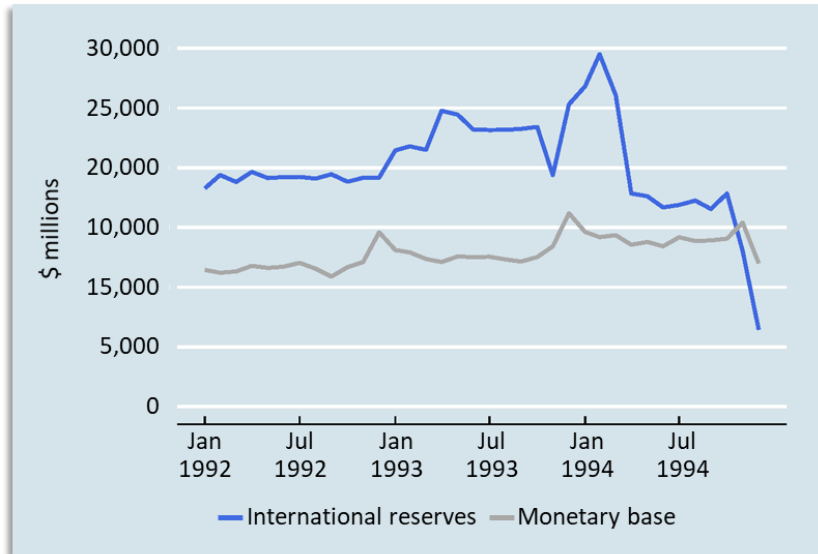
Foreign Exchange Market Intervention (4 of 6)

- Market (unsterilized) intervention can have a lasting effect on exchange rates.
 - Creates inflation in some nations and deflation in others.
 - E.g., previously, US and Eurozone wanted to maintain their current exchange rate e_0 .
 - The outcome is permanent, inflationary increase in the money supply in the Eurozone and deflationary decrease in the US.
 - If the resulting increase in Eurozone inflation and decrease in U.S. inflation were sufficiently large, the exchange rate would remain at e_0 without the need for further government intervention.
 - The money supply changes, and not the intervention itself, affect the exchange rate.

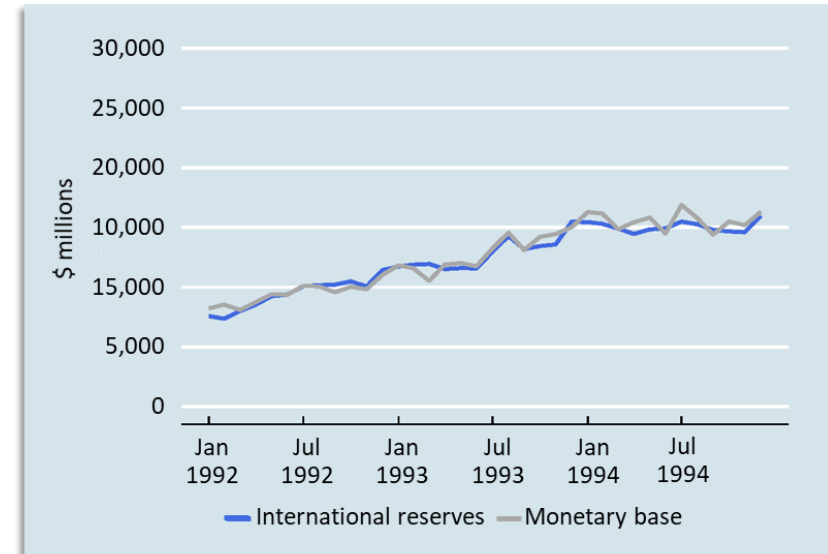
Foreign Exchange Market Intervention (5 of 6)

Example: Effects of Sterilized vs. Unsterilized Intervention

In 1994, Mexico sterilized its peso purchases by buying back government securities, causing its reserves to decline significantly while its money followed its usual growth pattern.



Conversely, Argentina's currency board prevented its ability to sterilize changes in reserves, forcing changes in Argentina's monetary base to closely match changes in its dollar reserves.



Foreign Exchange Market Intervention (6 of 6)

- **Example:** Currency interventions by the Swiss National Bank (SNB).
 - Between 2009-2010, the SNB made a sequence of four large currency interventions to weaken the value of the Swiss franc relative to the euro, with little long-term effect on the value of the franc.
 - Exchange rates appear to have been moved largely by basic market forces.

